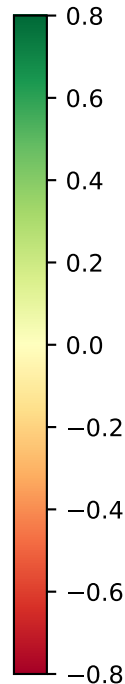


Stage 2 paired treatment effects: critic-free reward optimization is decisive

Sampled policy

Deterministic policy

Paired success change versus corrupted actor



Standard H=3

+16%

-2%

+8%

-36%

-16%

-18%

Standard H=6

+8%

-3%

-2%

-38%

-8%

-22%

Standard H=15

-4%

-5%

-13%

-40%

-18%

-16%

Standard H=20

+19%

-4%

+3%

-28%

-4%

-16%

Entropy-off H=3

+12%

-3%

-11%

-32%

-18%

-24%

Entropy-off H=6

+13%

-3%

-21%

-32%

-18%

-24%

Entropy-off H=15

+10%

-1%

-8%

-50%

-14%

-14%

Entropy-off H=20

-2%

-1%

-1%

-16%

-18%

-30%

Reward-only H=3

+22%

+56%

+0%

-2%

+64%

+36%

Reward-only H=6

+21%

+61%

+9%

-8%

+72%

+36%

Reward-only H=15

+28%

+90%

+16%

-2%

+78%

+28%

Reward-only H=20

+29%

+85%

+5%

+6%

+80%

+46%

Clean-policy distillation

+22%

+8%

+30%

-18%

+10%

+24%

Kill goblin

Bury bones

Chop logs

Kill goblin

Bury bones

Chop logs